

Bufotalin 预算耗尽版 Agentic Blackboard 报告

真实运行：本地 PDF 只在 scout 选中后进入流程，其余 action 全部走 controller/tool 链

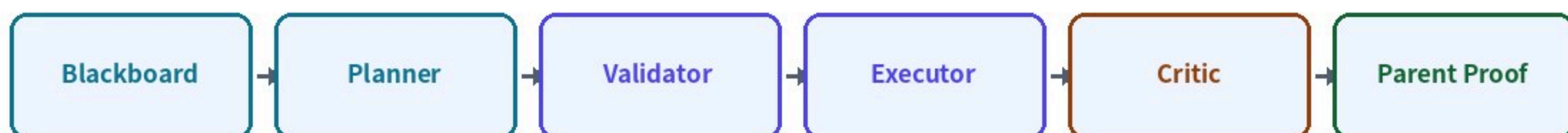
目标 bufotalin

SMILES 重原子 / 环数 32 / 5

最终结论 unresolved (child_solved_parent_unresolved)

停止原因 max_round_budget_exhausted, rounds 5/5

本地 PDF /root/autodl-tmp/AutoPlanner/1-s2.0-S0040402025001668-main.pdf (878443 bytes)



关键边界

本报告展示的是持续探索到轮次预算耗尽，不是提前 stop。即使过程中出现文献候选、类比排序、guided ChemEnzy 或子目标尝试，最终 solved 仍只由 stitched_parent_route_proof.v1 决定。

为什么这一轮不是直接停止

本次 planner 被配置为预算耗尽演示，除 proof 接受外不主动 stop

1. 实际停止原因：max_round_budget_exhausted。

2. 轮次预算：5/5，budget_exhausted=True。

3. action batch 每轮仍经过 schema、预算和 raw reaction 注入校验；planner 不能直接宣称 solved。

4. 本地 PDF 没有预先塞给视觉工具；只有 `search\_literature` 选择它以后，下一轮才执行 PDF 渲染。

5. 最后一轮尝试 parent proof/stitch。proof 不满足时，final verdict 保持 non-solved。

Action 决策轨迹

每个 action 都有 rationale、expected_artifact、success_condition 和执行记录

第 1 轮 - 生成目标侧断键假设 [accepted, useful=True] 理由：先从 bufotalin 目标本身生成可审计断键/桥接假设，避免直接把远端文献或类比路线当成证明。
第 1 轮 - 检索目标近端文献 [accepted, useful=True] 理由：blackboard 缺少目标近端文献证据；agent 因此请求文献 scout，本地 PDF 在此时作为候选源提供。
第 2 轮 - 渲染/索引本地 PDF 结构证据 [accepted, useful=True] 理由：scout 已经把本地 PDF 标为候选源；先把 PDF 渲染/索引为当前 run 的视觉证据。
第 2 轮 - 排序类比假设 [accepted, useful=True] 理由：已有目标侧假设，可以先排序类比探索方向；排序只影响策略，不作为 proof。
第 3 轮 - 视觉抽取文献结构链 [accepted, useful=True] 理由：PDF 结构证据已产生；尝试用真实视觉链 agent 从当前 PDF 图片抽取 source-detail 结构序列。
第 3 轮 - 编译 exact 文献行 [accepted, useful=False] 理由：视觉链若可用，则把 source-detail validated steps 编译为 exact rows；不可用时保留失败证据。
第 4 轮 - 运行 guided ChemEnzy [accepted, useful=True] 理由：已有 bridge tasks、类比排序和可能的 exact rows；执行一次真实 guided ChemEnzy，但结果仍必须过 verifier。
第 4 轮 - 扩展上游子目标 [accepted, useful=True] 理由：若 failure critic 或 exact rows 暗示高级终端/桥接子目标，则尝试上游子目标搜索；子目标 solved 不能升级父目标。
第 5 轮 - 拼接父路线证明 [accepted, useful=True] 理由：最后一轮尝试 deterministic parent-route proof；没有连通 proof 时只能保持 unresolved/partial。
第 5 轮 - 构建失败批判报告 [accepted, useful=True] 理由：预算耗尽前整理 guided/插件/verifier 失败，把原因转成下一轮可读 blackboard 状态。

文献与 PDF 证据链

agent 请求文献后，用户提供的本地 PDF 才进入抽取流程

Scout source candidates

- doi:10.1016/j.tet.2025.134610: local_pdf=True, recommendations=extract_pdf_literature_structures, extract_visual_literature_chain
- query:bufotalin:polycyclic_cage_core: local_pdf=False, recommendations=extract_pdf_literature_structures, extract_visual_literature_chain, compile_exact_literature_rows
- query:bufotalin:protecting_group_level_transformations: local_pdf=False, recommendations=extract_pdf_literature_structures, extract_visual_literature_chain, compile_exact_literature_rows

PDF render/index summary

- present=True, accepted=True, source=/root/autodl-tmp/AutoPlanner/1-s2.0-S0040402025001668-main.pdf
- counts={'compound_text_snippet_count': 0, 'indexed_image_count': 0, 'rendered_page_count': 6, 'scheme_crop_count': 0}
- reasons=无

Visual chain extraction

- present=True, accepted=False, status=completed
- candidate_step_count=1
- reasons=visual_literature_chain_extraction_gaps

真实工具执行摘要

这里列出的都是 controller 记录的 tool_calls，不是报告脚本伪造结果

Tool calls

- extract_pdf_literature_structures: status=accepted, accepted=True, elapsed=0.723s, reasons=无
- extract_visual_literature_chain: status=rejected, accepted=False, elapsed=420.276s, reasons=visual_literature_chain_extraction_gaps
- compile_source_detail_chain_route: status=rejected, accepted=False, elapsed=0.003s, reasons=no_compiled_downstream_assets, missing_one_step_row_for_product, no_chain_unrolled
- run_guided_chemenzy_rerun: status=accepted, accepted=True, elapsed=123.707s, reasons=高级同骨架终端被拒绝, 出现无法解释的大重原子跳跃, no_verifier_accepted_stock_closed_route
- run_route_expansion_subgoal_search: status=accepted, accepted=True, elapsed=61.841s, reasons=没有子目标被 verifier 证明闭合
- stitch_literature_chain_with_subgoal_route: status=rejected, accepted=False, elapsed=0.005s, reasons=literature_chain_not_accepted, literature_terminal_invalid_or_missing, literature_terminal_subgoal_target_mismatch, subgoal_route_not_solved, subgoal_verifier_not_accepted

Artifact result snapshot

- compiled_exact: present=True, accepted=False, reasons=['no_compiled_downstream_assets', 'missing_one_step_row_for_product', 'no_chain_unrolled']
- guided_chemenzy: present=True, accepted=False, route_status=fake_closed_rejected
- route_expansion: present=True, accepted=False, subgoals=1
- stitched_route: present=True, accepted=False, reasons=['literature_chain_not_accepted', 'literature_terminal_invalid_or_missing', 'literature_terminal_subgoal_target_mismatch', 'subgoal_route_not_solved', 'subgoal_verifier_not_accepted']

Blackboard 与最终门禁

失败、类比和 exact evidence 都只能进入状态；solved 需要 parent proof

Bridge tasks

- target_proximal_bridge: polycyclic_cage_core -> target-proximal cage intermediate with matched ring system
- target_proximal_bridge: protecting_group_level_transformations -> same-core alcohol/phenol protection-state analog
- target_proximal_bridge: analogue_diene_iminium_scaffold_inspiration -> target-proximal cage precursor, not a distant analogue product
- target_proximal_bridge_required: target_side -> Find a target-proximal intermediate that explains the large skeleton change.
- upstream_terminal_synthesis: advanced_same_scaffold_terminal -> Find upstream synthesis for rejected same-scaffold terminal.

Selected analogical hypotheses

- target_side_analogue_diene_iminium_scaffold_inspiration score=20 verify=target_proximal_bridge, deterministic_parent_proof
- target_side_analogue_diene_iminium_scaffold_inspiration score=20 verify=target_proximal_bridge, deterministic_parent_proof
- target_side_polycyclic_cage_core_preservation score=20 verify=core_retention_audit, max_unexplained_heavy_atom_jump, route_verifier_acceptance

Parent proof gate

- accepted=False, solved=False, route_status=child_solved_parent_unresolved
- reasons=子目标路线未连到父路线桥接点, exact 文献片段未连到父路线, 父路线 verifier 未接受, 库存审计未通过, unexplained_large_atom_jump
- final=unresolved, solved=False

审计文件与验证

报告输出、run artifacts 和 focused tests

测试：python -m pytest tests/test_agentic_blackboard_controller.py tests/test_failure_critic.py tests/test_parent_route_proof.py tests/test_open_research_experience.py tests/test_codex_entry_harness_contract.py -q: 139 passed, 2 skipped in 23.01s

Blackboard artifact refs

- bufotalin_budget_r1_generate_disconnection_hypotheses_target_side_disconnection_hypotheses_v1: /root/autodl-tmp/AutoPlanner/results/shared/bufotalin_agentic_blackboard_budget_exhaustion_valid_policy_20260609/bufotalin_budget_r1_generate_disconnection_hypotheses_target_side_disconnection_hypotheses_v1.json
- bufotalin_budget_r1_search_literature_literature_scout_report_v1: /root/autodl-tmp/AutoPlanner/results/shared/bufotalin_agentic_blackboard_budget_exhaustion_valid_policy_20260609/bufotalin_budget_r1_search_literature_literature_scout_report_v1.json
- bufotalin_budget_r2_extract_pdf_literature_structures_literature_pdf_structure_evidence_v1: /root/autodl-tmp/AutoPlanner/results/shared/bufotalin_agentic_blackboard_budget_exhaustion_valid_policy_20260609/bufotalin_budget_r2_extract_pdf_literature_structures_literature_pdf_structure_evidence_v1.json
- bufotalin_budget_r2_rank_analogical_hypotheses_analogical_hypothesis_ranking_v1: /root/autodl-tmp/AutoPlanner/results/shared/bufotalin_agentic_blackboard_budget_exhaustion_valid_policy_20260609/bufotalin_budget_r2_rank_analogical_hypotheses_analogical_hypothesis_ranking_v1.json
- bufotalin_budget_r3_compile_exact_literature_rows_compiled_source_detail_chain_route_v1: /root/autodl-tmp/AutoPlanner/results/shared/bufotalin_agentic_blackboard_budget_exhaustion_valid_policy_20260609/bufotalin_budget_r3_compile_exact_literature_rows_compiled_source_detail_chain_route_v1.json
- bufotalin_budget_r3_extract_visual_literature_chain_visual_literature_chain_extraction_result_v1: /root/autodl-tmp/AutoPlanner/results/shared/bufotalin_agentic_blackboard_budget_exhaustion_valid_policy_20260609/bufotalin_budget_r3_extract_visual_literature_chain_visual_literature_chain_extraction_result_v1.json
- bufotalin_budget_r4_expand_child_target_route_expansion_subgoal_search_result_v1: /root/autodl-tmp/AutoPlanner/results/shared/bufotalin_agentic_blackboard_budget_exhaustion_valid_policy_20260609/bufotalin_budget_r4_expand_child_target_route_expansion_subgoal_search_result_v1.json
- bufotalin_budget_r4_run_guided_chemenzy_guided_chemenzy_rerun_result_v1: /root/autodl-tmp/AutoPlanner/results/shared/bufotalin_agentic_blackboard_budget_exhaustion_valid_policy_20260609/bufotalin_budget_r4_run_guided_chemenzy_guided_chemenzy_rerun_result_v1.json

Artifact bundle keys

- agent_blackboard, analogical_hypothesis_ranking, compiled_downstream, compiled_downstream_payload, failure_critic_report, guided_chemenzy, literature_pdf_structure_evidence, literature_pdf_structure_evidence_dir, literature_scout, parent_route_proof, route_expansion_subgoal_search, route_failure_feedback, source_detail_chain_route, stitched_semisynthesis_route, visual_literature_chain_extraction, visual_structure_candidate_chain